ACS Energy LETTERS
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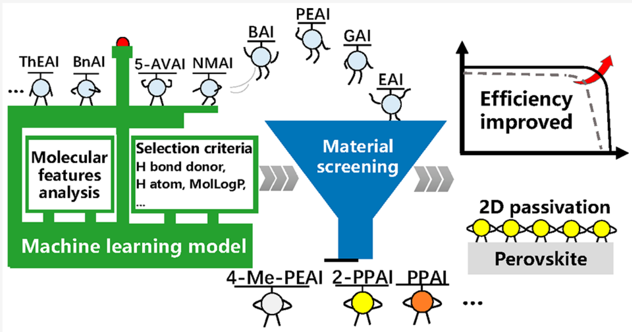
Machine-Learning-Assisted Screening of Interface Passivation Materials for Perovskite Solar Cells

Chongyang Zhi, Suo Wang, Shijing Sun, Can Li, Zhihao Li, Zhi Wan, Hongqiang Wang,* Zhen Li,* and Zhe Liu*

Cite This: *ACS Energy Lett.* 2023, 8, 1424–1433 Read Online

ABSTRACT: **Interface passivation using an ammonium salt can effectively improve the power conversion efficiency (PCE) of perovskite solar cells (PSCs). Despite significant PCE improvement achieved in previous studies, the selection criteria for ammonium salts are not fully understood. Here we apply a machine-learning (ML) method to investigate the relationship between the molecular features of ammonium salts and the PCE improvement of PSCs. We establish an ML model using an experimental data set of 19 salts to predict the PCE improvement after passivation. Three molecular features (hydrogen bond donor, hydrogen atom, and octane–water partition coefficient) are identified as the most important features of selecting an ammonium salt for passivation. The ML model is further used to screen ammonium salts from a pool of 112 salts in the PubChem database. FAMACs and FAMA-based PSCs fabricated with a model-recommended salt (2-phenylpropane-1-aminium iodide) achieve PCEs of 22.36% and 24.47%, respectively.**



Hybrid organic–inorganic perovskite solar cells (PSCs) are some of the most promising candidates for next-generation low-cost photovoltaics. Thus far, the power conversion efficiency (PCE) of PSCs has already surpassed 25%, closing the gap to record efficiency for silicon solar cells.[1−4] Among the several approaches used to improve solar cell performance, suppressing nonradiative carrier recombination by surface/interface passivation is one of the key factors for the recent progress in the efficiency increase of PSCs.[5−12] Recently, two-dimensional (2D) perovskites have been utilized as capping layers to passivate the surface of the perovskite absorber layer.[3,13−23] Because the 2D capping layer is formed by a reaction of ammonium halide solutions with a 3D perovskite absorber, the perovskite photovoltaics (PV) community refers to this type of device as a 2D/3D PSC.[24−26] The current record PCE achieved by these 2D/3D perovskite solar cells is 25.7% using octylammonium iodide (OAI).[1] Other studies have also demonstrated a relative efficiency enhancement of 10–20% using different ammonium salts. Examples of commonly used ammonium salts include ethylammonium iodide (EAI), *n*-butylammonium iodide (*n*-BAI), and phenethylammonium iodide (PEAI).[27−29]

Despite the efforts of the perovskite PV community in device engineering, it remains challenging to establish a clear selection criterion for numerous ammonium salts to achieve good surface passivation and device PCE. Machine learning (ML) has been proposed in recent studies to assist in material screening with minimal experimental input data. Hartono et al.[30,31] systematically studied the impact of various molecular features on the stability of perovskite thin films with 2D capping layers in a humid environment. A good materials screening approach was established by machine learning for the 2D perovskite layer as a humidity barrier. A similar approach was adopted by Yu et al.,[32] who investigated the tendency of ammonium salts to form a 2D perovskite phase on a $MAPbI_3$ surface. All these previous studies focused on the screening of ammonium salts for 2D perovskite formation and film stability. A machine-learning model should be also used to predict the passivation effect and PCE enhancement in 2D/3D PSCs at the device level. Therefore, to better understand the

Received: December 12, 2022
Accepted: February 1, 2023
Published: February 15, 2023

https://doi.org/10.1021/acsenergylett.2c02818
ACS Energy Lett. 2023, 8, 1424–1433

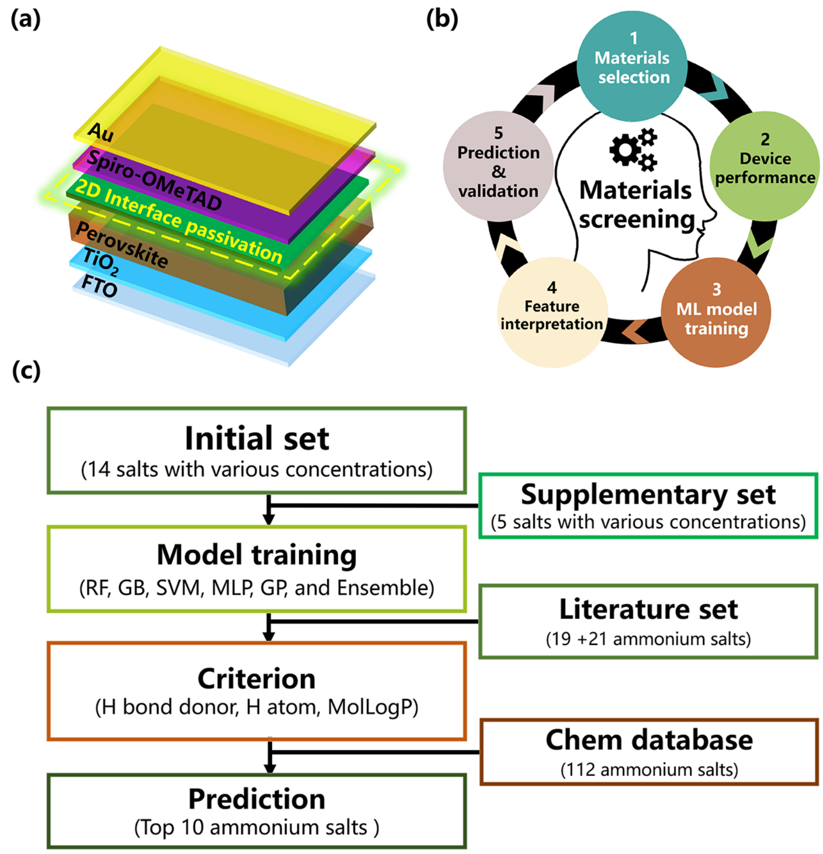


Figure 1. (a) The device structure of a 2D/3D perovskite solar cell with a 2D perovskite capping layer. (b) Flowchart of machine-learning-assisted screening concept. (c) Schematic diagram of the actual working procedure of ammonium salt screening and feature analysis in this work.

passivation effect of the 2D capping layer, the relationship between the molecular features and power conversion efficiency (PCE) should be thoroughly investigated.

In this study, we adopt a data-driven machine-learning analysis to establish the most important molecular features that affect the PCE of a device. We demonstrate an ML framework for ammonium iodides to passivate $(FA_{0.83}MA_{0.17})_{0.93}Cs_{0.07}Pb(I_{0.83}Br_{0.17})_3$-based (FAMACs) triple-cation perovskites. This study collected an initial representative selection of 19 ammonium iodides (with varying precursor concentrations) to build a machine-learning regression model, including 14 initial data sets and 5 additional explorative data selected by Latin Hypercube Sampling (LHS). The regression model correlates the 13 specific molecular features to PCE improvement ratios, which allows us to identify the most important molecular features by a Shapley Additive Explanations (SHAP) analysis. The SHAP analysis can establish a suitable range for the most important molecular features to achieve a high PCE. We use the chemical property criteria to downselect 10 suitable candidates out of the 112 ammonium iodides available in the PubChem database. To validate the model prediction, we conduct new experiments on 6 different ammonium salts. Among them, we achieve PV device PCEs of 22.36% and 24.47% for FAMACs-based and FAMA-based PSCs with 2-phenylpropan-1-aminium iodide (2-PPAI), which has not been used for perovskite passivation before.

The 3D FAMACs-based PSCs were prepared using an antisolvent method with the structure of FTO/c-TiO_2/mp-TiO_2/FAMACs-based perovskite/spiro-OMeTAD/Ag, as shown in Figure 1a. The 2D perovskite passivation layers were spontaneously formed by a reaction between the ammonium iodide salts and the 3D perovskite, as illustrated in Figure S1. Ammonium salts were experimentally investigated as passivation layers to improve the device PCE. Ammonium iodide salts with varying cation sizes and functional groups (i.e., alkyl chains and aromatic rings) have been demonstrated as promising passivation layers to improve efficiency.[25,26,33−35]

The flowchart in Figure 1b presents the procedures used to investigate the relationships between the molecular features of the 2D ammonium salts and PCE of the FAMACs-based PSCs. The general concept of a workflow for ML-assisted material screening is divided into the following five steps: material selection, device performance, ML model training, feature interpretation, and prediction and validation. First, based on existing literature and our chemical knowledge, we compiled a list of ammonium salts that can form 2D perovskites and passivate the PSCs. Second, these ammonium salts were used to fabricate 2D/3D PSCs at various concentrations (four examples are shown in Figure S2). The performance parameters of the devices were collected using standard $J-V$ curve measurements. Third, the molecular structures and chemical properties of these ammonium salts were transformed into feature descriptors and the PCE improvement ratio of the corresponding 2D/3D PSCs was loaded into the algorithms to build a regression model. Finally, after the appropriate ML regression model was trained, it was used to predict the potential of other ammonium salts in the chemical database.

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ACS Energy Lett. 2023, 8, 1424−1433

The predicted salts were used to fabricate PSCs to validate the material screening criteria.

The screening steps used in this study are shown in Figure 1c. (i) Initially, 14 commonly used ammonium salts with various concentrations (31 data points from 14 salts) were collected. (ii) To avoid data set bias the initial data set was supplemented with 5 salts (15 data points including concentrations) selected using the LHS method (more details are discussed in Supporting Information) to increase the sampling diversity. (iii) A correlation between the feature descriptors of the ammonium salts and the device PCE improvement ratio was established using the ML regression model. (iv) After establishing the ML regression model, we added 21 ammonium salts to the database obtained from existing studies and built a Shapley Additive Explanations (SHAP) framework to determine the importance ranking of the material descriptors based on their contributions to the device efficiency. (*v*) Finally, we predicted top-10 and bottom-10 ammonium salts from a PubChem database and used some of the predicted salts for experimental validation.

Before building an ML regression for ammonium salts and device PCEs, we need to indicate the characteristic features of ammonium salts as quantitative descriptors.[36−38] In this study, RDKit was used in this work to compute the material descriptors for the ammonium salts. Fourteen characteristic descriptors were chosen to describe the molecular features of the salts, with their definitions being given in Table 1. The descriptors include compositional features, such as C, H, N, F, and O counts, and chemical structural property descriptors, such as the molecular weight, topological molecular polar surface area (TPSA), number of hydrogen-bond donors/acceptors, complexity index, and the molecular lipid−water partition coefficient (molecular LogP or MolLogP), etc.

Table 1. Definition of the 14 Descriptors for the Ammonium Salts in This Study

descriptor	definition	descriptor	definition
C atoms	no. of carbon atoms	H-bond donor	no. of hydrogen-bond donors
H atoms	no. of hydrogen atoms	H-bond acceptor	no. of hydrogen-bond acceptors
N atoms	no. of nitrogen atoms	Rotatable bond	no. of rotatable bonds
F atoms	no. of fluorine atoms	Complexity	topological complexity index
O atoms	no. of oxygen atoms	Ipc	information for polynomial coefficients of a molecular graph
molecular weight	weight of the molecule in g/mol	MolLogP	molecular lipid−water partition coefficient
TPSA	topological molecular polar surface area	heavy atom	no. of atoms except for hydrogen

To visualize the distribution of the material's descriptors in the two-dimensional plane, we perform a principal component analysis (PCA) to transform the descriptors onto the two principal components:[39] e.g., PCA-1 and PCA-2. The detailed procedure can be found in section 2c in the Supporting Information. Figure 2a demonstrates the feature distributions of the material candidates after the dimension reduction of the 14 feature descriptors to the two principal components. The initial data set contains 14 ammonium salts (red symbols in Figure 2). These materials were slightly concentrated in the two-dimensional PCA space, which is typical for researcher-driven data sets. However, an ML model trained with a biased data set can have less generality and may poorly predict in regions with no existing data. To improve the diversity of the training data set, we supplemented 5 additional ammonium salts (blue symbols in Figure 2) using the LHS method. These 5 ammonium salts cover the remaining region of the material feature space more diversely. The gray symbols represent the ammonium salts in the PubChem database, which were not used to build the machine-learning model. There was no apparent overlap among the 14 existing samples and 5 additional samples in the PCA plot, indicating that the supplement data increase the variety of materials.

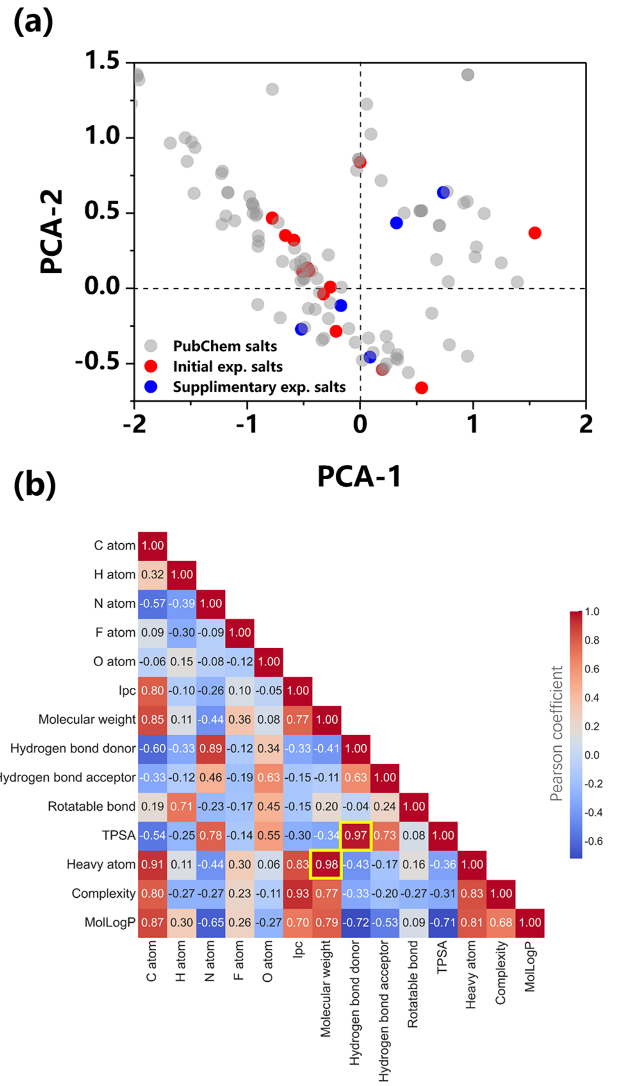

Figure 2. A two-dimensional representation of the feature descriptors for ammonium salts is given by using the principal component analysis algorithm. (a) The 14 ammonium salts in the historical data set are marked in red. The 5 additional selections of ammonium salts are marked in blue. The gray symbols represent the ammonium salts in the expanding data set. (b) Pearson correlation matrix for the 14 selected descriptors of the 19 ammonium salts.

The Pearson correlation heat map of the data set shown in Figure 2b was used to check for strong correlations between any pair of descriptors. To improve the accuracy of an ML model, highly correlated descriptors need to be eliminated.

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ACS Energy Lett. 2023, 8, 1424−1433

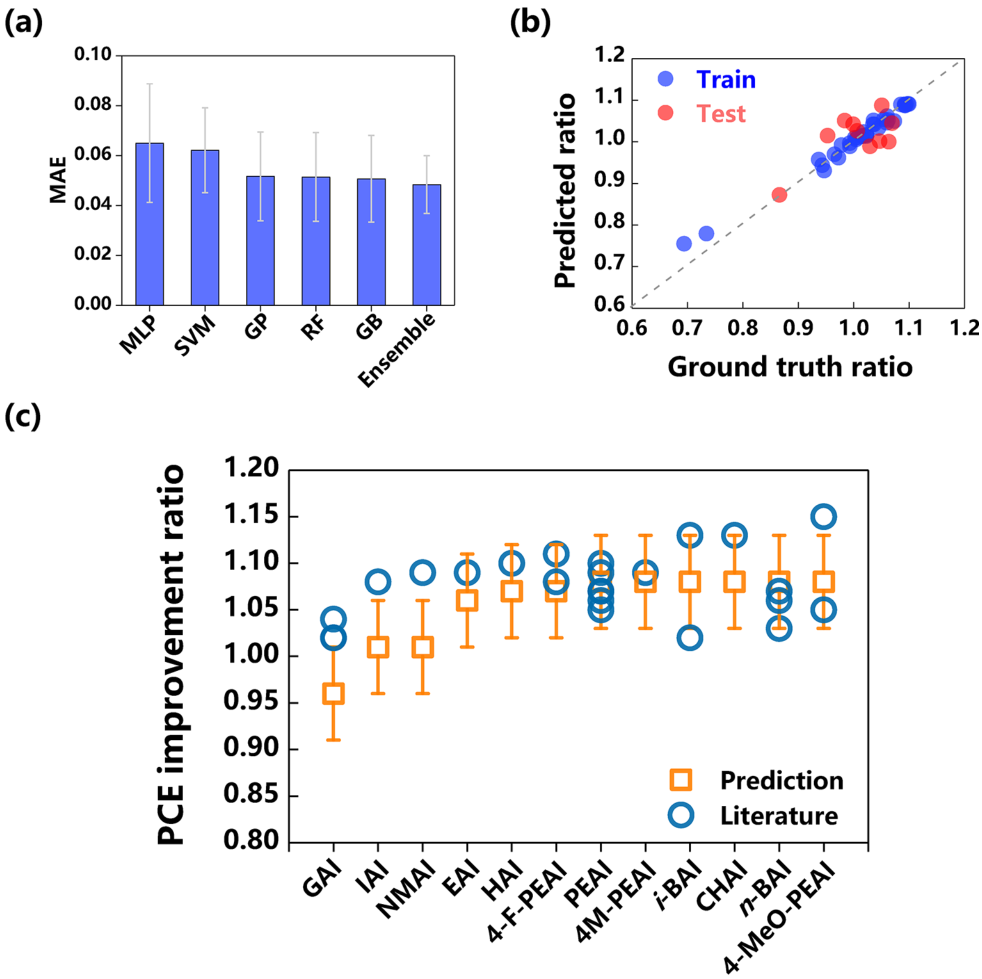

Figure 3. Result of ML model training. (a) The test group MAE values of the different regression models. The error indicates the standard deviation of the MAE in the 200 repetitive runs of the train-test-data split and model training. (b) Prediction vs ground truth plot for the ensemble model. (c) Comparison of the prediction and literature PCE enhancement ratio with the same device architecture.[27,40−55]

Pearson correlation coefficients (PCCs) were calculated for each pair of descriptors to quantify the linear correlation. Only one of the descriptor pairs with Pearson correlation coefficients larger than 0.95 was selected for ML model training. For instance, TPSA was filtered out owing to its high correlation with hydrogen bond donors (PCC = 0.97). The heavy atom was filtered because of a 0.98 correlation with the molecular weight. Finally, the input variables to the ML model are in 13 dimensions, including the remaining 12 molecular feature descriptors, along with a processing variable, the concentration of the precursor solution with ammonium salts. The target variable is the maximum PCE improvement ratio, which was defined as the ratio of the maximum PCEs of passivated devices over control devices. The input and target variables of our experimental data set are given in Table S1.

Several popular regression algorithms have been used to build an effective prediction model, such as random forest (RF) regression, gradient boosting (GB) regression, support vector machine (SVM) regression, multilayer perceptron (MLP) regression, and Gaussian process (GP) regression.[30−32,36−39] In addition to individual regression models, we also proposed an ensemble learning approach to integrate five individual ML regressors. The ensemble learning model combines the outputs from all regression models in order to eliminate model bias and combine the advantages of regression with different data-processing architectures (e.g., tree-based RF model and kernel-based SVM model), which reduces the possibility of overfitting in our case of a small training data set. The RF, GB, SVM, MLP, and GP regression models were used as the primary base learners, and the prediction results were integrated by a secondary meta-learner using a new RF regression model. The molecular descriptors with varying precursor concentrations were used as the input variables, and the corresponding PCE improvement ratio was used as the output/target variable.

To evaluate the accuracy of the model, the data set was randomly divided into training and testing sets with a ratio of 4:1. The model hyperparameters were tuned in a random search with the 5-fold cross-validation method. The hyperparameters of these models are shown in Table S2. Owing to our small data set, we obtained the average and standard deviation of the prediction MAEs by running 200 times. The MAE error for the regression models is summarized in Figure 3a. The full list of MAEs for the training and testing sets can be found in Table S3. We found that the ensemble learning model demonstrated better prediction results with a lower MAE of 0.0484 ± 0.0152. The ensemble learning model combines the advantage of each regression algorithm and reduces model biases. The output of the model prediction versus the ground truth for the ensemble model is shown in Figure 3b.

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ACS Energy Lett. 2023, 8, 1424−1433

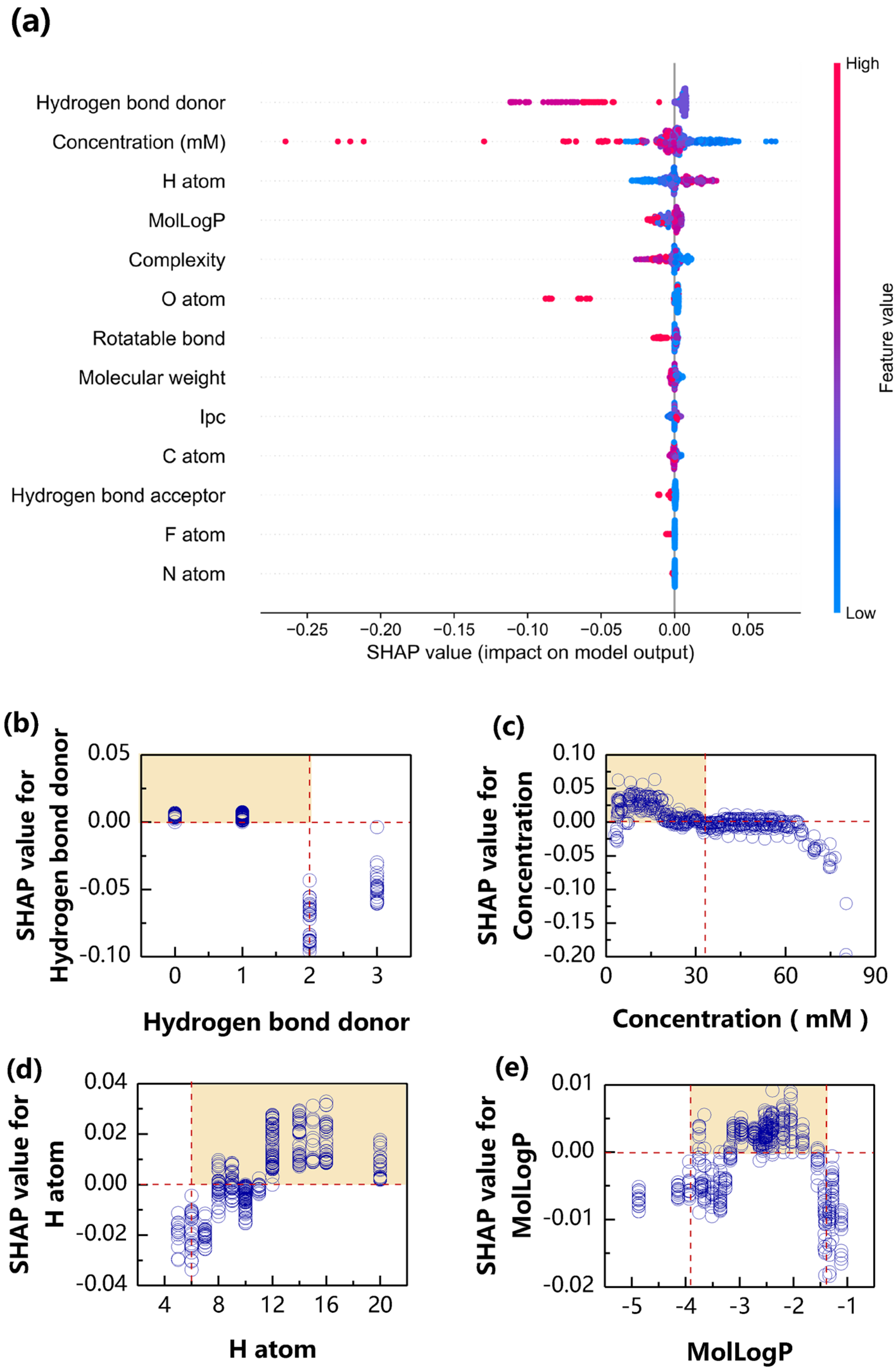

Figure 4. SHAP analysis of the ML ensemble regression model prediction. (a) Importance ranking of different features. The SHAP value indicates positive or negative contributions to the device's efficiency. Red corresponds to a high value of the features, whereas blue corresponds to a low value of the features. (b−e) Relationship between the important descriptors of the ensemble model and their SHAP values: (b) Hydrogen bond donor; (c) precursor concentration; (d) hydrogen atom; and (e) molecular lipid−water partition coefficient (MolLogP).

We further collected the PCE improvement ratios of certain ammonium iodides from the literature data set[27,40−55] and compared them with our model predictions, as shown in Figure 3c. The predicted results sufficiently agree with the reported results in the literature. The result demonstrates the effectiveness of the prediction results derived from our machine-learning model.

After the ML ensemble model was established, feature importance analysis become particularly helpful for identifying the most influential input variables.[31,38,56,57] The contribution to the PCE improvement ratio is presented by the SHAP values, as shown in Figure 4a. The 13 input variables (i.e., 12 material descriptors and precursor concentration) are ranked based on their overall absolute impact on efficiency, regardless of the positive or negative contributions. The hydrogen bond donor is considered the most important feature connected with efficiency in our model. The second in ranking is the concentration of ammonium salts. This is in agreement with our experience that the concentration of the ammonium salt needs to be optimized to achieve the best performance in

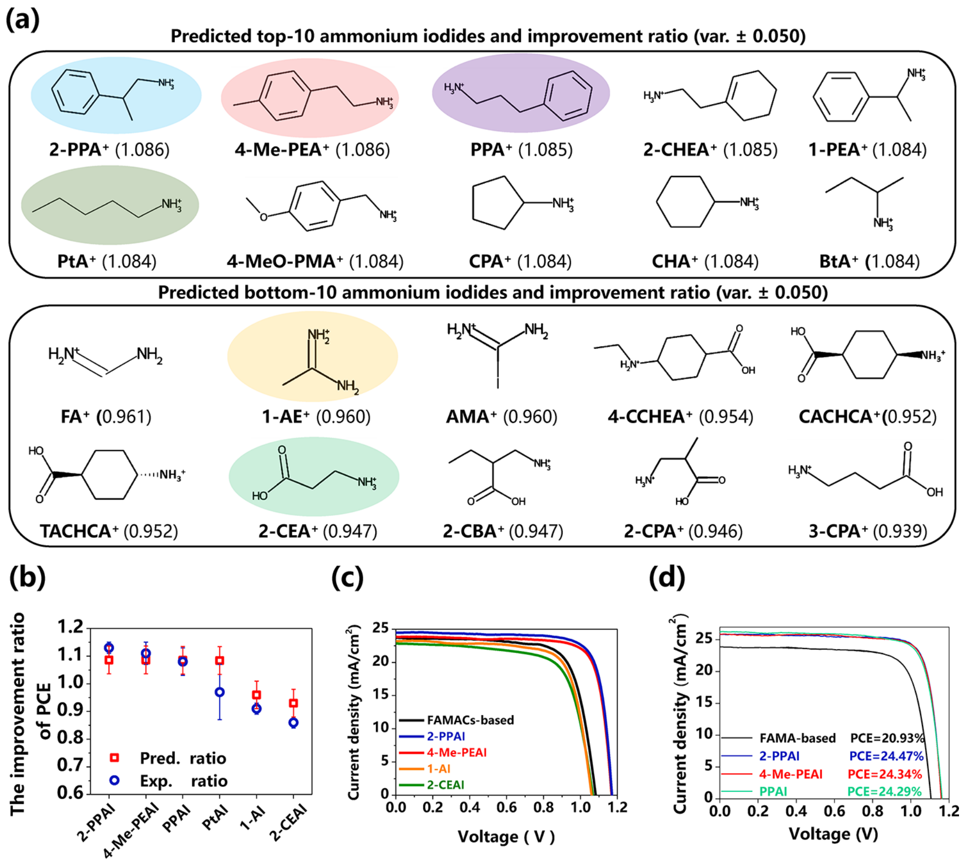

Figure 5. Experimental validation of machine learning models: (a) top-10 and bottom-10 ammonium salts predicted from the ensemble model; (b) PCE improvement ratios of the tested salts; (c) *J*–*V* curves of FAMACs-based solar cells with 2-PPAI, 4-Me-PEAI, PPAI, PtAI, 1-AEI, and 2-CEAI treatments. (d) *J*–*V* curves of FAMA-based solar cells treated with the 3 best salts predicted by our model.

PSCs. Among the remaining 11 descriptors, H atom and MolLogP have lower ranks but also play important roles in the model.

Figure 4b–e helps us visualize the SHAP values versus the values of the four most important features. Figure 4b demonstrates the effect of the hydrogen bond donor on the device efficiency. It was found that the SHAP value becomes negative when hydrogen bond donors are >1. A hydrogen bond donor greater than 1 represents the electronegative atoms (N, O, and F) that can form hydrogen bonds with the MA and FA cations in the perovskite. These hydrogen bond donors increase the reactivity of the ammonium salt with the 3D perovskite,[31] leading to excessive 2D-toward transformation.[58,59] The effects of concentration agree with our understanding that a low concentration means insufficient passivation and a high concentration leads to a thick passivation layer that hinders carrier transport. The H atom (number of hydrogen atoms) is correlated with the size and conjugated degree of the ammonium. More H atoms indicate a larger molecule, keeping the ammonium at the interface rather than replacing the A-site cations (FA/MA/Cs) in the perovskite film, which is beneficial for interface passivation. H counts over 16 start to show deterioration of the device performance due to increased resistance of the bulky ammonium. The MolLogP value positively correlates with the hydrophobicity of the molecule.[60] We found that a suitable MolLogP value delivers a beneficial effect to the PCE, indicating the benefits of a hydrophobic interface layer not only to the humidity stability but also to the efficiency of PSCs. However, a MolLogP that is too high affects the wettability of the film when depositing spiro-OMeTAD, leading to a poor coverage spiro-OMeTAD layer and shunting paths in the resulting device. At this stage, the model provides a set of rules for screening ammonium salts with the potential to improve efficiency through surface passivation. To improve the efficiency, the selected candidates should have the following features but are not limited to these: (a) hydrogen bond donor <2, (b) $8 \leq$ H atom ≤ 20, and (c) $-3.8 \leq$ MolLogP ≤ -1.4.

The ensemble model was then used to predict the PCE improvement of 112 ammonium salts collected from the PubChem database.[61] The full name of salts is shown in Table S4, and the predicted PCE improvement ratios are arranged from highest to lowest. An ammonium iodide with a ratio of greater than 1 is referred to as a "good salt", while the opposite is considered a "bad salt". Top-10 and bottom-10 ammonium iodides from the ML prediction and their improvement ratios of PCE are shown in Figure 5a. Their molecular feature values are shown in Tables S5 and S6, respectively. It is encouraging to find that several salts from the top-10 ammonium iodides were shown to greatly improve the efficiency of PSCs in recent reports.[40−42]

To test the accuracy of the model's prediction, we selected ammonium iodides from both the good and bad salt groups for model validation. 2-Phenylpropan-1-aminium iodide (2-PPAI), 2-(*p*-tolyl ethan-1-aminium iodide (4-Me-PEAI), phenylpropan-1-aminium iodide (PPAI), and pentylammonium iodide (PtAI) were selected from the good salt group. 1-Aminoethan-1-iminium iodide (1-AEI) and (2-carboxy)-

https://doi.org/10.1021/acsenergylett.2c02818
ACS Energy Lett. 2023, 8, 1424−1433

ethylammonium iodide (2-CEAI) were selected from the bad salt group. The statistical analysis of PCE and performance parameters of the 6 groups of FAMACs-based devices are shown in Figure S3 and Table S7. The best device was obtained by the 2-PPAI treatment (also the best in prediction), which achieved a PCE of 22.36%. The 1-AI- and 3-CEAI-treated devices both demonstrated lower PCE than the control devices, with PCEs of 17.98% and 17.13%, respectively. A comparison between the experimental and model-predicted PCE improvement ratios of the tested salts is shown in Figure 5b. The value distribution of the PCE improvement ratio in experiments tallies with that in prediction; however, an outlier still exists. This shows the limitation now that the present model is capable of predicting improvement trends instead of the accurate value due to the small initial data set. The typical *J*−*V* curves of treated devices are shown in Figure 5c.

To demonstrate the generality of our model prediction, the top 3 salts were further applied to FAMA-based PSCs. The *J*−*V* curves of the FAMA-based control and 2-PPAI-treated, 4-Me-PEAI-treated, and PPAI-treated devices are shown in Figure 5d and Figure S4a−c. The best PCEs of 20.93%, 24.47%, 24.34%, and 24.28% are achieved in PSCs without treatment and with 2-PPAI, 4-Me-PEAI, and PPAI treatment, respectively. The performance parameters of the FAMA-based control and salt-treated devices are compared in Table S8. The maximum power point (MPP) tracking is shown in Figure S4d−f. Following 2-PPAI, 4-Me-PEAI, and PPAI surface passivation, the stabilized output is improved from 19.98% to 24.17%, 24.01%, and 23.78%, respectively.

In summary, we demonstrated the use of an ML ensemble regression model for screening ammonium salts to form 2D perovskite layers for interface passivation in PSCs. Inspired by the recent initiative of data-centric ML for materials science, we tackle a common critical challenge that a typical researcher-driven data set contains unbalanced sampling. We proposed the LHS method to add new data in the underexplored regions in the material feature space. Using a relatively diverse data set of 19 ammonium salts, the ML regression model established a numerical correlation between the molecular features of ammonium salts and their improvement in the power conversion efficiency of PSCs. To reduce the model bias and enhance the robustness of the model, we chose an ensemble learning strategy with five base learners of different model architectures. The model predictions agreed with the results in recently published scientific literatures.

In addition, using a Shapley value analysis, the ML model suggested screening criteria for the molecular features of potential ammonium salts to achieve efficient PSCs. Namely, hydrogen bond donor, hydrogen atom, and lipid−water partition coefficient (MolLogP) should be primarily considered when selecting new ammonium salts to achieve a good interface passivation effect. Among the suggested materials by the ML model, 2-PPAI showed significant improvements in PCE, leading to device PCEs of up to 22.36% and 24.47% for the FAMACs-based and FAMA-based PSCs, respectively. We believe that ML-assisted material screening can help us interpret the crucial molecular features for perovskite interface passivation and effectively predict the material candidates. The proposed ML workflow can be easily adapted for screening a wide range of organic molecules to improve the performance and stability of optoelectronic devices.

■ ASSOCIATED CONTENT

Data Availability Statement

The data and code necessary to reproduce this work can be downloaded at a GitHub repository (github.com/chrisliu2007/passivation_layer_screening).

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsenergylett.2c02818.

Experimental methods, material details, analysis data of the *J*−*V* curvea, coefficients for the regression model, machine-learning algorithm functions with the Scikit-learn package, and the entire data set (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Hongqiang Wang − *School of Materials Science and Engineering, Northwestern Polytechnical University, Xi'an, Shaanxi 710072, People's Republic of China; State Key Laboratory for Solidification Processing, Northwestern Polytechnical University, Xi'an, Shaanxi 710072, People's Republic of China; NPU-Chongqing Innovation Center, Chongqing 401147, People's Republic of China;* orcid.org/0000-0003-1262-1958; Email: hongqiang.wang@nwpu.edu.cn

Zhen Li − *School of Materials Science and Engineering, Northwestern Polytechnical University, Xi'an, Shaanxi 710072, People's Republic of China; State Key Laboratory for Solidification Processing, Northwestern Polytechnical University, Xi'an, Shaanxi 710072, People's Republic of China;* orcid.org/0000-0003-1177-2818; Email: lizhen@nwpu.edu.cn

Zhe Liu − *School of Materials Science and Engineering, Northwestern Polytechnical University, Xi'an, Shaanxi 710072, People's Republic of China; State Key Laboratory for Solidification Processing, Northwestern Polytechnical University, Xi'an, Shaanxi 710072, People's Republic of China; NPU-Chongqing Innovation Center, Chongqing 401147, People's Republic of China*; Email: zhe.liu@nwpu.edu.cn

Authors

Chongyang Zhi − *School of Materials Science and Engineering, Northwestern Polytechnical University, Xi'an, Shaanxi 710072, People's Republic of China; State Key Laboratory for Solidification Processing, Northwestern Polytechnical University, Xi'an, Shaanxi 710072, People's Republic of China*

Suo Wang − *School of Materials Science and Engineering, Northwestern Polytechnical University, Xi'an, Shaanxi 710072, People's Republic of China; State Key Laboratory for Solidification Processing, Northwestern Polytechnical University, Xi'an, Shaanxi 710072, People's Republic of China*

Shijing Sun − *Department of Mechanical Engineering, Massachusetts Institute of Technology, Cambridge, Massachusetts 02139, United States;* orcid.org/0000-0002-6179-1390

Can Li − *School of Materials Science and Engineering, Northwestern Polytechnical University, Xi'an, Shaanxi 710072, People's Republic of China; State Key Laboratory for Solidification Processing, Northwestern Polytechnical*

https://doi.org/10.1021/acsenergylett.2c02818
ACS Energy Lett. 2023, 8, 1424−1433

University, Xi'an, Shaanxi 710072, People's Republic of China

Zhihao Li − *School of Materials Science and Engineering, Northwestern Polytechnical University, Xi'an, Shaanxi 710072, People's Republic of China; State Key Laboratory for Solidification Processing, Northwestern Polytechnical University, Xi'an, Shaanxi 710072, People's Republic of China*

Zhi Wan − *School of Materials Science and Engineering, Northwestern Polytechnical University, Xi'an, Shaanxi 710072, People's Republic of China; State Key Laboratory for Solidification Processing, Northwestern Polytechnical University, Xi'an, Shaanxi 710072, People's Republic of China*

Complete contact information is available at: https://pubs.acs.org/10.1021/acsenergylett.2c02818

Author Contributions

C.Z. and S.W. contributed equally to this work. Conceptualization: Z. Li, Z. Liu, S.S., C.L., H.W. Coding: S.W., Z. Liu. Experiments: C.Z. Manuscript writing: C.Z., Z. Liu, Z. Li with help and revision feedback from H.W., C.L., S.S., Z. Li, Z.W. Project Coordination: Z. Liu, Z. Li, C.L., H.W.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

The authors acknowledge fruitful discussions with Prof. Qian Ye and Dr. Pengfei Guo from NPU. The authors acknowledge funding support from the National Key R&D Program of China (Grant No. 2019YFB1503201), the National Natural Science Foundation of China (Grant Nos. 51902264, 52172238, 52103286, and 52102304), the Natural Science Foundation of Shaanxi Province (Grant No. 2020JM-093), the Shaanxi Bureau of Science and Technology (Award No. 2022KWZ-07), and the Fundamental Research Funds for the Central Universities (Grant No. 3102019JC0005).

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https://doi.org/10.1021/acsenergylett.2c02818
ACS Energy Lett. 2023, 8, 1424−1433